

Requirement Analysis Phase

Technology Stack Architecture & Component Specifications

Attribute	Project Information
Project Title	Auto Ticket Classification using Flow Designer
Phase / Stage	2. Requirement Analysis
Document Title	Technology Stack (Architecture & Stack)
Platform & Scope	ServiceNow Utah / Vancouver / Washington DC (Global Scope)
Team ID	SWTID-2026-4988

1. Technical Architecture Overview

The Auto Ticket Classification system is architected entirely on the ServiceNow Enterprise Cloud Platform. It utilizes a 4-tier model comprising Presentation, Workflow Automation, Data Persistence, and Communications. This architecture ensures high availability, strict multi-tenant security, horizontal scalability, and zero external dependency footprint.

2. Table-1: Components & Technologies

S.No	Component / Layer	Functional Description	ServiceNow Core Technology / Module
1	Presentation Layer (User Interface)	Provides intuitive incident submission and agent triage views; manages dynamic form rendering and client interactions.	ServiceNow Form Designer / Form Builder / Next Experience UI Framework
2	Process Automation Layer (Workflow)	Evaluates record trigger criteria, parses short descriptions for keywords, executes conditional branches, and updates records.	ServiceNow Flow Designer (Workflow Studio) - No-Code Flow Engine
3	Data Model & Persistence Layer	Stores incident records, manages auto-numbering counters, maintains referential integrity, and enforces dependent choice lists.	ServiceNow Relational Database Layer (MariaDB / MySQL Architecture) on u_incident_workflow
4	Notification & Messaging Engine	Generates dynamic HTML email notifications, resolves Caller email addresses via data pills, and handles SMTP outbound delivery.	ServiceNow Email Engine (sys_email / System Logs > Emails) with SMTP Relay
5	Lifecycle & Deployment Subsystem	Tracks, packages, and promotes configuration records across instances (Development -> QA -> Production) in standardized XML.	ServiceNow System Update Sets (Local Update Sets > Export to XML)

3. Table-2: Platform Specifications & Environment Details

Specification Parameter	Configured Value & Environment Standard	Architectural Rationale
Host Cloud Platform	ServiceNow Enterprise SaaS Cloud (Instance: Personal Developer Instance / Enterprise Node)	Guarantees 99.99% availability, automated data redundancy, and automated backups.
Application Scope	Global Application Scope	Ensures seamless access to platform tables sys user and sys user group without cross-scope restrictions.
Platform Release Version	Utah / Vancouver / Washington DC / Xanadu Release	Leverages latest Workflow Studio and Flow Designer builder features.
Database Table Name	u_incident_workflow (Label: Incident Workflow)	Isolated custom table dedicated to academic IT incident triage workflows.
Auto-Numbering Schema	Prefix: INC Starting Number: 500 Digits: 5 (e.g. INC00509)	Ensures unique, human-readable identifier standard aligned with ITIL incident conventions.
Dependent Choice Engine	Dictionary Choice Specification: Dependent on Category	Maintains relational integrity between Category and Subcategory values.
Email Protocol	Outbound SMTP over TLS / SSL	Secure, auditable email notification dispatch tracked in system logs.

4. Technical Architecture Diagram (Textual Representation)

[Caller: Student/Teacher] --> Submits Incident (Caller + Short Description)

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[Custom Table: u_incident_workflow] --> Generates INC005xx Auto-Number

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[Flow Trigger: Category is empty] --> Evaluates Record Created

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+---> [Branch 1: "Wi-Fi" / "Network"] ==> Set Network / Wi-Fi

+---> [Branch 2: "Projector"] ==> Set Hardware / Projector

+---> [Branch 3: "Forgot password"] ==> Set Access / Forgot Password

+---> [Branch 4: "Slow computer"] ==> Set Performance / Slow Computer

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[Action: Update Record] --> Commits Category & Subcategory to Database

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[Action: Send Email] --> Dispatches Acknowledgment via sys_email to Caller